

UAV DESIGN**VI Semester**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE23	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
UNIT-I	INTRODUCTION TO UAV							
Historical Development, UAV Classifications, Successful UAVs , Design Criteria, Objectives, and Priorities, Design Process, Systems Engineering Approach, UAV Design Steps,UAV Conceptual Design, UAV Preliminary Design, UAV Detail Design.								
UNIT-II	AERODYNAMIC DESIGN OF WING AND TAIL							
Introduction, Fundamentals of Aerodynamics, Wing Design Procedure, Airfoil Selection/Design, Wing Design Technique, Wing Design Steps, Tail Design, Design Procedure, Tail Configuration, Horizontal Tail Design Technique, Horizontal Tail Parameters, Tail Planform Area and Tail Arm, Tail Airfoil Section, Vertical Tail Design Parameters, Vertical Tail Location, Vertical Tail Moment Arm (lvt), Vertical Tail Design Technique								
UNIT-III	FUSELAGE DESIGN							
Introduction, Fuselage Design Fundamentals,Fuselage Aerodynamics,Autopilot Compartment,Optimum Length-to-Diameter Ratio,FuselageLofting,Fuselage Design Steps, fuselage configurations of UAVs.								
UNIT-IV	ROTARY-WING UAVS AND HYBRID UAVS							
Introduction, Rotary-Wing Configurations, Aerodynamics Single Rotor,Twin Co-axial Rotors,Twin Tandem Rotors,Multicomputer,aerodynamicsof Tilt Rotor,Tilt Wing,Fixed-Wing Quad copter Combination aerodynamics								
UNIT-V	VTOL UAVs							
Introduction VLOTL UAVs, - Quad copters-components-Design principles-Aerodynamics-Control-Hex copters-aerodynamics-control								
Text Books:								
1. Paul Fahlstrom, Thomas Gleason - Introduction to UAV Systems-Wiley (2012) 2. Reg Austin, "Unmanned Air Systems: UAV Design, Development and Deployment", First Edition, Wiley Publishers, 2015.								
Reference Books:								
1. Mirosaw Adamski, "Power units and power supply systems in UAV", New Edition, Taylor and Francis Group publishers, 2014. 2. Skafidas, "Microcontroller Systems for a UAV", KTH, TRITA-FYS 2002:51 ISSN 0280-316X. 34, 2002.								
COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:								
1. Identify and analyze the various subsystems comprising UAVs, exploring their functionalities and interactions within the overall UAV system.								

2. Evaluate various types of UAV wings and their impact on performance, including flapping wings, rotary wings, and their contribution to climbing, range, endurance, and gliding flight.
3. Analyze the structural aspects of UAVs, examining the types of loads, materials, construction techniques, and their significance in ensuring structural integrity.
4. Explore the mission planning and control station for UAVs, categorizing types, physical configurations, planning, navigation, and the interfaces of the Mission Planning and Control System (MPCS)
5. Apply the principles of system design and selection across the conceptual, preliminary, and detailed design phases in the context of UAVs.